π-SA: Precision Intelligent Surgical Agent

The problem of fully autonomous steering guidewire and catheter in endovascular procedures

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THE PROBLEM

A floppy wire, a life to be saved

1.9 M

neurons lost every minute in a typical untreated large-vessel ischemic stroke

Time is brain: every minute spent navigating counts.

Thin, flexible devices, experts far away

Full autonomy is now an industry goal

ARPA-H awarded Siemens Healthineers up to **$31.1 M** to develop autonomous remote endovascular robotics for stroke (Aug 2026).

Saver, “Time is brain — quantified”, Stroke 2006 · Yao et al., arXiv 2504.15327

STARTING POINT · HCMR-EV

A full stack, built for deployment

INPUT
Operator or policy
device commands

→

ROS 2
OpenNG3
command arbitration

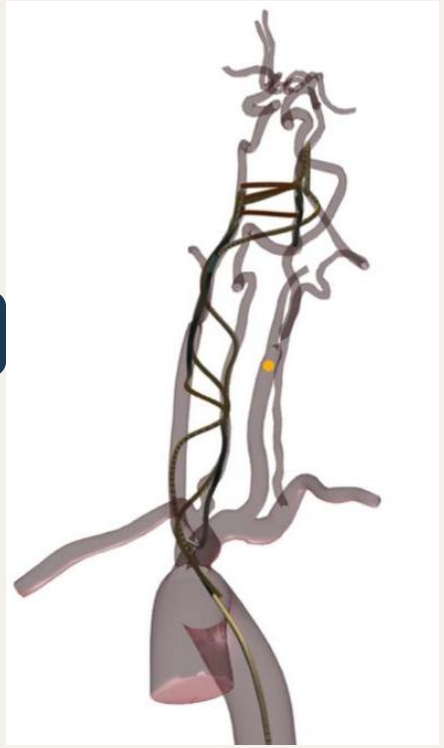
→

VM · UDP
Monadnock
generated robot dynamics

→

BROWSER
EndoNav
JavaScript physics + WebGL

HCMR-EV-RL: online SAC running inside this stack, plus ACT / SmolVLA training scripts. Simulator, RL and policies already existed.



BECAUSE IT TARGETS A REAL SYSTEM, ITS RL HAD:

- **Noisy inputs only**
- **Real time, 5 Hz**
- **Euclidean reward**
- **Fixed goals**

Starting point: HCMR-EV project — OpenNG3, Monadnock model, EndoNav, HCMR-EV-RL, LeRobot recorder

WHAT WE BUILT

Our proposal

01 · SIMULATE

Xcath (our new simulator)

- Real vessel mesh
- Cosserat rods (XPBD): guidewire + catheter
- Robot actuator model: delay + filter
- **Gymnasium env**, 22 in parallel

02 · TEACH

SAC teacher

- One expert per vessel
- Privileged state: **353 values** incl. **forces**
- **Geodesic** reward, goal = tree node
- **Curriculum learning** + randomisation

03 · RECORD

LeRobot

- Teacher rollouts recorded at 20 Hz
- **Fluoroscopy-style** renderer, 480×480
- **LeRobot** v3.0, Open-H schema

04 · IMITATION

SmolVLA student

- 450 M parameters
- In: **image + text + robot state**
- Out: next 50 **robot controls**
- Trained on successful episodes only

Vessel in → expert demonstrations out → **a policy that steers from an X-ray image and a sentence**

THE KEY IDEA

The RL policy is not the product. It is the teacher.

A teacher only has to produce good demonstrations. So it may use information no real sensor provides, and it doesn't have to live inside the slow real-time stack.

Teacher · sees everything · 0.7 M parameters → **Student** · sees what an operator sees · 450 M parameters

π-SA

01 · SIMULATE · XCATH

Our improved simulator

Teacher Policy
reads the state, sends 4 commands: insert + rotate for guidewire and catheter

→

Actuator dynamics model
Monadnock simplified in python, parameters from system identification

→

Rod physics
XPBD Cosserat rods, guidewire inside catheter, vessel-wall containment

→

Contact and forces
wall and lumen friction, force taxonomy of Pantano et al.

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22 Environments in parallel on one machine: 44h simulated in 8h

0 browsers, ROS nodes or VMs in the loop: a headless NumPy simulator behind a Gymnasium API

Checked against the original
validated against the page's own JavaScript, same physical model

Macklin et al., XPBD, MIG 2016 · Kugelstadt & Schömer, Cosserat rods, SCA 2016 ·

02 · TEACH · SOFT ACTOR-CRITIC (SAC)

A privileged expert that sees everything

OBSERVATION · 353 VALUES

- Goal + path ahead along the centreline
- Distal segments of both devices: position relative to the tip, velocity, rotation
- Distance to the vessel wall, per segment
- **Every force term, both devices**: contact, wall friction, viscous drag, elastic, wire–catheter friction, handle axial + torque, tip, per segment
- Actuator state + previous actions

→

ACTOR π(a | s)

Policy network · 0.7 M parameters

stochastic, maximum-entropy

CRITICS Q(s, a)

Value of each action

trained off-policy from a replay buffer

REWARD

progress along the centreline; success = geodesic distance < 5 mm + force penalty

→

4 COMMANDS

Insert + rotate

guidewire and catheter, mm/s and rad/s

↓

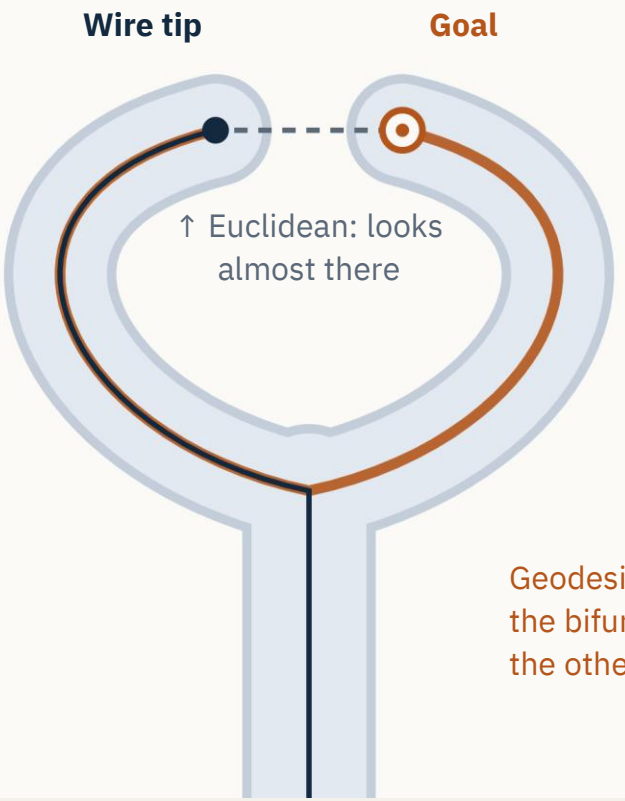
Actuator model → physics

with realistic robot lag

The teacher knows *why* it is stuck: pressing on the wall, rubbing the catheter, or dragging through blood.

02 · TEACH · REWARD

A smarter reward: Distance measured along the vessel



BEFORE · EUCLIDEAN

Straight-line distance from tip to goal. In a branching vessel, a point in the wrong branch can look almost reached.

OURS · GEODESIC

Distance along the centreline tree extracted from the vessel mesh: the length the wire really has to travel.

< 5 mm

geodesic distance to the goal counts as success

Prior RL for guidewire navigation: Karstensen et al., IJCARS 2022

03 · RECORD

From teacher rollouts to a VLA dataset



One frame	Content
Image	480×480 fluoroscopy-style frame
State	30 values
Action	4 commands, in mm/s and rad/s
Outcome	success or failure

Cadene et al., LeRobot, 2024 · Same Open-H schema as the original HCMR-EV recorder

04 · VISION-LANGUAGE-ACTION

SmolVLA learns from pixels and values

INPUTS

- **Image** · fluoroscopy-style, 480×480
- **Task text** · “Navigate to goal”
- **Robot state** · 30 values

→

VISION-LANGUAGE BACKBONE

SmolVLM

image, text and state become one token sequence

→

ACTION EXPERT

Flow matching

attends to the backbone’s features and turns noise into an action chunk

→

OUTPUT

Next 50 commands

4 device commands per step: insert + rotate, wire and catheter

Teacher · 0.7 M parameters

Student · 450 M parameters

sees only what an operator sees

Real-time control

action chunking: The inference time is hidden inside the motion.

Shukor et al., SmolVLA, arXiv 2025 · Lipman et al., Flow matching, ICLR 2023 · Black et al., π0, arXiv 2024

THANK YOU

Questions Time

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